

AppliTrack – Electric Appliance Management System

Backend Features List for Diploma Final Year Project

- 1. User Management:
 - - User Registration & Login (with JWT Authentication)
 - - Role-based Access: User, Admin, Technician
- 2. Appliance Management:
 - - Add/Edit/Delete Appliances
 - - Track details: Model, Serial Number, Purchase Date, Warranty, Usage History, Service History
- 3. Usage Tracking:
 - - Log appliance usage (date, duration, energy consumed)
- 4. Automated Maintenance & Warranty Alerts:
 - - Cron Jobs to check warranty expiry & service schedule
 - - Notifications via Email/SMS
- 5. Inventory Management:
 - - Manage appliance stock
 - - Track appliance status (working, under repair, discarded)
- 6. Complaints Management System (Unique Feature):
 - - Users submit complaints (e.g., 'TV not working')
 - - Store: User ID, Appliance ID, Description, Date-Time, Status (Open/In Progress/Resolved)
 - - Admin assigns to Technician
 - - Technicians receive notifications
 - - Track complaint status
 - - Store full complaint history
- 7. Notifications System:
 - - Email/SMS notifications (Nodemailer/Twilio)
 - - Store notification history
- 8. Reports & Analytics:
 - - Reports: Total appliances, Energy usage trends, Warranty expirations, Complaint history & resolution times
- 9. Audit Log System:
 - - Log actions (user login, appliance CRUD, complaints created/resolved)
- 10. API Documentation:
 - - Swagger/OpenAPI documentation for all endpoints
- 11. Backup & Restore (Optional):
 - - Export/Import appliance & complaint data
- Extra Future Scope (Optional):
 - - Predictive Maintenance based on usage patterns
 - - IoT Integration for real-time monitoring

This list provides a comprehensive backend structure that makes AppliTrack unique and suitable as a final year diploma project.